INTERNSHIP TASK 01

Strategic Planning and Data Exploration in Logistics

A Python-based proposal for delivery performance and delay analysis

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REPORT PURPOSE

This report defines a realistic logistics analytics problem, proposes measurable performance indicators, reviews public data and technical resources, and outlines a feasible Python workflow. It is a strategic plan; no company-specific findings or model performance results are claimed.

Executive Summary

Logistics operations depend on reliable delivery schedules, efficient transportation, and visibility into shipment performance. Late deliveries can increase operating costs, disrupt customer plans, and make resource allocation harder. This project proposes a structured analysis of shipment records to understand where delays occur and which shipment characteristics are associated with delivery performance.

The proposed workflow uses Python with Pandas for data preparation, Matplotlib for visual exploration, and scikit-learn for an optional predictive baseline. Four core KPIs will be calculated where the dataset contains the necessary fields: on-time delivery rate, delay rate, average delay duration, and average shipping cost. Regression or classification may be considered after the data and target variable are validated; clustering can be used to group similar shipment profiles. Route optimization is treated as a future extension because it requires route, distance, vehicle-capacity, and operational-constraint data.

1. Background and Business Context

A logistics provider manages shipments across different regions, shipping modes, product categories, and delivery schedules. Operations teams may know that late deliveries occur, but may not have a consolidated view of how often they happen, whether some shipping modes or regions perform differently, or how actual shipping time compares with scheduled shipping time.

The World Bank's Logistics Performance Indicators provide a broader context for logistics performance. Their indicators consider dimensions such as supply-chain speed, connectivity, tracking, and reliability. Country-level indicators do not directly diagnose a single company's operations, but they reinforce the importance of measuring logistics outcomes consistently [1].

This proposed project narrows the problem to shipment-level delivery performance. The intended audience is a logistics operations manager who needs clear evidence to prioritize process improvements.

2. Project Definition

2.1 Problem statement

The company needs a repeatable, data-driven method to measure shipment timeliness, identify patterns associated with delays, and compare delivery performance across available operational categories. Without a consistent KPI baseline, improvement efforts may be based on isolated incidents rather than systematic evidence.

2.2 Aim

To design a Python-based logistics analysis workflow that converts historical shipment records into understandable KPIs, visual insights, and practical recommendations for delivery planning.

2.3 Objectives

- Identify a suitable public dataset and document its source, available fields, and limitations.
- Inspect, clean, and validate shipment data, including missing values, duplicate records, and data types.
- Calculate at least three logistics KPIs using transparent definitions.
- Explore differences in delivery performance by region, shipping mode, or other available categories.
- Assess whether a simple predictive model or clustering analysis is feasible and useful.
- Translate findings into practical recommendations and identify further data needed for route or capacity optimization.

2.4 Scope and assumptions

IN SCOPE	OUT OF SCOPE FOR WEEK 1

Planning the analytical workflow	Deploying a production logistics system
Public shipment data and Python examples	Accessing confidential company data
KPI definitions and exploratory analysis plan	Claiming improvements before analysis
Feasibility assessment for modeling	Live GPS tracking or real-time route optimization

3. Literature and Data Research

3.1 Public dataset candidate

A practical starting point is the DataCo Supply Chain Dataset, publicly listed on Kaggle. A project repository describing its use reports approximately 180,519 records and 53 columns, including actual and scheduled shipping days, delivery status, shipping mode, region, and a late-delivery-risk field [2, 3]. The dataset should be inspected directly before analysis, because column names, definitions, availability, and access conditions must be confirmed from the downloaded files and dataset page.

Dataset page: DataCo Supply Chain Dataset (Kaggle)

Supplementary column overview: Late Delivery Risk Prediction project

The DataCo dataset is a candidate, not a verified local file in this report. Before using it, confirm the download works, review its license and data dictionary, and check that the required fields are present. If access is restricted or the fields differ, select another public dataset with documented delivery time or status fields.

3.2 Research observations

- Operational data is more useful when planned and actual outcomes can be compared. A field representing actual shipping days and another representing scheduled shipping days can support a transparent delay measure.
- The World Bank's LPI 2.0 describes a move toward indicators grounded in shipment-level operational data and emphasizes supply-chain speed, connectivity, and reliability [1]. This supports the project's focus on measurable delivery outcomes.
- Pandas provides a standard read_csv function for importing CSV files into a DataFrame, making it suitable for initial data loading and inspection [4].
- Scikit-learn's LinearRegression estimates a linear relationship by minimizing residual sum of squares. It can serve as a simple baseline when a continuous target and appropriate input features are available [5].

3.3 Data fields to verify

Field / field group	Use in analysis	Validation needed
Actual shipping days	Observed shipment duration	Numeric, non-negative, plausible values
Scheduled shipping days	Expected duration / comparison baseline	Numeric, non-negative, consistent units
Delivery status / late-delivery risk	Cross-check and optional classification target	Understand exact definition and label encoding
Shipping mode	Compare service categories	Normalize spelling and category labels
Order region / market	Compare geographical groups	Check missing and inconsistent labels
Shipping cost field	Estimate average cost where available	Confirm currency, units, and whether it is per order or item

4. Key Performance Indicators (KPIs)

The KPI definitions below are proposed for the project. The final calculation must match the data dictionary and business rules. For example, a shipment is treated as on time when actual shipping days are less than or equal to scheduled shipping days, provided both fields represent comparable durations.

KPI	Definition / formula	Decision value
On-time delivery rate	Shipments where actual days ≤ scheduled days ÷ eligible shipments × 100	Shows the proportion meeting the schedule.
Delay rate	Shipments where actual days > scheduled days ÷ eligible shipments × 100	Shows the share requiring attention.

Average delay (days)	Mean of max(actual days − scheduled days, 0) for delayed shipments	Estimates delay severity among late shipments.
Average shipping cost	Sum of valid shipping costs ÷ records represented by those costs	Supports cost monitoring; confirm cost granularity.

KPI governance: report the number of eligible records and missing values beside each percentage. Exclude records with unavailable or invalid comparison fields from the relevant denominator, and state the exclusion rule. Do not treat cancelled or undelivered orders as on-time deliveries unless the business definition explicitly requires it.

5. Data Science Methods and Proposed Use

Method	Proposed application	Feasibility / caution
Exploratory Data Analysis (EDA)	Summarize distributions, missingness, outliers, delay rates by region and shipping mode.	First method to apply; findings are descriptive, not proof of causation.
Regression	Estimate actual shipping days or delay duration from appropriate shipment features.	Requires a meaningful numeric target, enough usable records, and leakage-free features.
Classification (optional)	Estimate late-delivery risk as a binary outcome.	Use only if the label is well-defined and available before the delivery outcome.
Clustering	Group shipments with similar duration, cost, shipping mode, or operational characteristics.	Scale numeric inputs and encode categorical inputs; clusters need business interpretation.
Optimization (future extension)	Explore route or vehicle allocation to reduce travel time or cost under capacity constraints.	Requires route distances, stops, vehicle capacities, time windows, and operational constraints.

5.1 Model validation strategy

- Create a baseline before applying complex methods. Compare model results with a simple benchmark.
- Split data into training and testing sets before fitting a model; use a time-based split if the data is chronological and the goal is future prediction.
- For regression, consider Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
- For classification, consider precision, recall, F1-score, and a confusion matrix; accuracy alone can hide poor performance on delayed shipments.
- Check for data leakage. Do not use information that would only be known after delivery to predict whether a delivery will be late.
- Document assumptions, limitations, and model performance. Do not promise that a model will eliminate delays.

6. End-to-End Strategic Roadmap

Phase	Activities	Output / checkpoint
1. Define	Confirm business question, scope, KPI definitions, and intended audience.	Approved project brief
2. Collect	Download public CSV files and read dataset documentation and license.	Source record and raw-data copy
3. Profile	Inspect shape, columns, types, missingness, duplicates, ranges, and category counts.	Data quality summary
4. Clean	Standardize names and categories; convert types; handle missing/invalid values using documented rules.	Analysis-ready dataset
5. Explore	Calculate descriptive statistics and plot delays by available categories.	EDA charts and initial observations
6. Measure	Calculate KPIs with clear denominators and exclusions.	KPI table / dashboard-ready summary
7. Model	If justified, train and validate a simple regression or classification baseline; optionally test clustering.	Validation metrics and model limitations
8. Recommend	Translate patterns into testable actions and identify data gaps.	Prioritized recommendations
9. Communicate	Present methodology, charts, assumptions, and next steps.	Report and presentation-ready summary

6.1 Process flow

Business problem and KPI definition

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Public data collection and documentation review

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Data profiling and quality checks

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Cleaning and feature preparation

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Exploratory analysis and KPI calculation

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Optional model training and validation

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Interpretation, recommendations, and reporting

7. Python Code Illustrations

The following snippets illustrate the proposed approach. They are templates, not a completed analysis. Confirm the actual column names in the downloaded CSV and adjust them before execution. Install dependencies if needed with: pip install pandas matplotlib scikit-learn

7.1 Load and inspect data

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("DataCoSupplyChainDataset.csv", encoding="latin1")

print("Shape:", df.shape)
print(df.head())
print(df.info())
print(df.isna().sum().sort_values(ascending=False).head(15))
print(df.describe(include="all").T.head(20))
```

Purpose: load the CSV, inspect its structure, identify columns with missing values, and review summary statistics. If the file uses a different encoding or filename, adjust the read_csv arguments.

7.2 Clean essential fields and calculate KPIs

```
# Verify these names against df.columns before running.
actual_col = "Days for shipping (real)"
scheduled_col = "Days for shipment (scheduled)"

for col in [actual_col, scheduled_col]:
    df[col] = pd.to_numeric(df[col], errors="coerce")

eligible = df[[actual_col, scheduled_col]].dropna()
eligible = eligible[
    (eligible[actual_col] >= 0) & (eligible[scheduled_col] >= 0)
].copy()

eligible["delay_days"] = (
    eligible[actual_col] - eligible[scheduled_col]
).clip(lower=0)

eligible["is_delayed"] = (
    eligible[actual_col] > eligible[scheduled_col]
)

total = len(eligible)
on_time_rate = (
    (~eligible["is_delayed"]).mean() * 100 if total else 0
)
delay_rate = (
    eligible["is_delayed"].mean() * 100 if total else 0
)
avg_delay_when_late = (
    eligible.loc[eligible["is_delayed"], "delay_days"].mean()
)

print(f"Eligible shipments: {total}")
```

```
print(f"On-time rate: {on_time_rate:.2f}%")
print(f"Delay rate: {delay_rate:.2f}%")
print(f"Average delay among late shipments: {avg_delay_when_late:.2f} days")
```

Interpretation: this example defines delay from actual versus scheduled shipping days. It should be used only after confirming that the two fields are comparable and that the dataset's business definitions match this rule. A missing average delay means there were no delayed records in the eligible subset or no valid delay observations.

7.3 Compare delay by shipping mode

```
# Run only if the listed category column exists.
if "Shipping Mode" in df.columns:
    temp = df[[ "Shipping Mode", actual_col, scheduled_col ]].copy()
    temp[actual_col] = pd.to_numeric(temp[actual_col], errors="coerce")
    temp[scheduled_col] = pd.to_numeric(temp[scheduled_col], errors="coerce")
    temp = temp.dropna(subset=[actual_col, scheduled_col])
    temp["is_delayed"] = temp[actual_col] > temp[scheduled_col]

    delay_by_mode = temp.groupby("Shipping Mode")["is_delayed"].mean() * 100
    print(delay_by_mode.sort_values(ascending=False))

    delay_by_mode.sort_values(ascending=False).plot(kind="bar")
    plt.title("Delay Rate by Shipping Mode")
    plt.ylabel("Delayed shipments (%)")
    plt.xlabel("Shipping mode")
    plt.tight_layout()
    plt.show()
```

Purpose: compare delivery performance across shipping modes. Differences can guide further investigation, but they do not establish that shipping mode alone caused the delay.

7.4 Optional regression baseline

```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, root_mean_squared_error

# Use only features available at prediction time.
numeric_features = ["Order Item Quantity"]
categorical_features = ["Shipping Mode", "Market"]

required = numeric_features + categorical_features + [actual_col]
model_df = df[[c for c in required if c in df.columns]].copy()

# Ensure all selected fields exist before fitting.
if all(c in model_df.columns for c in required):
    model_df[actual_col] = pd.to_numeric(model_df[actual_col], errors="coerce")
    model_df = model_df.dropna(subset=[actual_col])
    X = model_df[numeric_features + categorical_features]
    y = model_df[actual_col]

    preprocessor = ColumnTransformer([
        ("num", Pipeline([
```

```
            ("imputer", SimpleImputer(strategy="median")),
            ("scale", StandardScaler())
        ]), numeric_features),
        ("cat", Pipeline([
            ("imputer", SimpleImputer(strategy="most_frequent")),
            ("encode", OneHotEncoder(handle_unknown="ignore"))
        ]), categorical_features)
    ])

    model = Pipeline([
        ("prep", preprocessor),
        ("regressor", LinearRegression())
    ])

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
    model.fit(X_train, y_train)
    predictions = model.predict(X_test)
    print("MAE:", mean_absolute_error(y_test, predictions))
    print("RMSE:", root_mean_squared_error(y_test, predictions))
```

This optional baseline is illustrative and should be run only if all listed fields exist and are suitable. In real forecasting, prefer a time-aware split when appropriate and remove any feature that leaks information from the delivery outcome. If the selected dataset does not contain these fields, revise the feature list or omit this model.

8. Expected Outcomes and Decision Impact

Because the dataset has not been analyzed in this planning report, the outcomes below are expected deliverables rather than observed findings.

- A documented dataset profile that shows the available fields, data-quality issues, and limitations.
- A reproducible KPI summary that quantifies on-time performance and delays according to stated definitions.
- Charts highlighting where delay rates differ across available categories such as shipping mode or region.
- A feasibility decision on whether regression, classification, or clustering is justified by the available data.
- Evidence-based recommendations for further investigation, such as reviewing poor-performing segments, improving exception monitoring, or collecting missing operational data.
- A list of additional fields needed for future route optimization, including distances, route stops, vehicle capacities, time windows, and cost constraints.

8.1 How results may support decisions

Potential observation	Possible management response
One region has a higher delay rate	Investigate regional constraints, handoffs, and carrier performance before changing resources.
One shipping mode has longer delivery times	Compare service promises, shipment mix, and operational context; review mode allocation.
Shipping costs vary across comparable shipment groups	Check cost definitions and investigate contracts, distance, package size, and service level.
A predictive baseline identifies higher-risk shipments	Pilot exception alerts for operational review; monitor false alarms and model drift.

9. Risks, Limitations, and Ethics

- Public datasets may be synthetic, anonymized, incomplete, or not representative of a specific company's operations.
- Column definitions and dataset licensing must be checked before use; cite the source and respect its terms.
- Missing data and inconsistent category labels can distort KPI results if not handled transparently.
- Historical associations do not prove causation. Recommendations should be tested with operational teams.
- Predictive models can be misleading if they use information unavailable at decision time or if the target label is defined using the same features.
- Shipment datasets may include location or customer-related fields. Avoid exposing personal information in the report and use only data needed for the analysis.

10. Conclusion

This strategic plan proposes a feasible first-stage logistics data science project focused on delivery performance and delay analysis. The project begins with a clearly defined business question, uses public shipment data subject to verification, and measures performance through transparent KPIs. Python tools support data inspection, cleaning, visualization, and optional predictive analysis.

The most important outcome of Week 1 is a well-defined and reproducible analytical plan. Once the dataset is validated and the workflow is executed, the resulting evidence can help logistics teams prioritize investigations, improve performance monitoring, and determine whether more advanced forecasting or optimization is warranted. Any operational improvement should be confirmed through follow-up measurement rather than assumed in advance.

References

1. [1] World Bank. Logistics Performance Indicators (LPI 2.0), About and methodology. Accessed 23 October 2026. https://lpi.worldbank.org/en/about
2. [2] Kaggle. DataCo Supply Chain Dataset. Dataset listing; review current access and license before use. https://www.kaggle.com/datasets/jolenechen/datacosupplychaindataset
3. [3] Aldhaferi-Mahmood. Late Delivery Risk Prediction, GitHub repository. Used as a supplementary description of dataset fields and workflow. https://github.com/Aldhaferi-Mahmood/Late-Delivery-Risk-Prediction
4. [4] pandas documentation. pandas.read_csv. CSV import reference. https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html
5. [5] scikit-learn documentation. LinearRegression. Model definition and usage reference. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

Submission checklist

- Replace the placeholders on the cover page with your details.
- Download and inspect the dataset; verify the file name and column names before running code.
- Check the dataset license and ensure the report accurately describes the source.
- If you execute the code, add genuine output tables or charts and describe only results you observed.
- Review the document against your internship organization's submission instructions.